

UTKARSH SARAOGI

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MSCS student with 3 years of experience at HPE and hands-on work across batch and real-time pipelines, distributed systems, and cloud infrastructure; building toward DE roles where scalable data systems and pipeline reliability all matter.

EDUCATION

Northeastern University, Boston | MS in Computer Science

Sep.2025 - Present

Coursework: MLOps, NLP, Data Mining, Database Management Systems, Algorithms

Manipal Institute of Technology, Manipal | BTech in Electronics and Communication

May 2022

Minor Specialization: Data Science and Data Modelling

TECHNICAL SKILLS

Data Engineering : Kafka, Airflow, dbt, Snowflake, Apache Spark, ETL/ELT, Delta Lake
Database Systems : MySQL, MongoDB, PostgreSQL
Programming & Core CS : Python, SQL, Java, PySpark, System Design
Cloud & DevOps : AWS (S3, Lambda), GCP (GCS, BigQuery), Docker, Git, CI/CD, Distributed Systems
Data Science & ML : Scikit-learn, PyTorch, Pandas, Matplotlib, A/B Testing, Causal Inference, LLM Interpretability

PROFESSIONAL WORK EXPERIENCE

Software Engineer (Cloud Developer)

Aug. 2022 – Aug. 2025

Hewlett Packard Enterprise

Bangalore, India

- Monitored data consistency across distributed telemetry flows spanning **15+** microservices ingesting switch metrics (VLAN/VXLAN segment data, hardware health signals); reduced MTTR by **40%** through systematic root cause analysis on upstream payload failures and data quality anomalies.
- Designed batch-parallel execution framework distributing data flow validation scripts across switch fleets simultaneously; reduced end-to-end test cycle times by **55%** through workload partitioning and optimized batch scheduling.
- Owned deployment lifecycle for data pipeline services using Jenkins CI/CD; implemented automated validation gates enforcing data contract integrity across distributed services before production promotion.
- Built structured training dataset from **4,000+** unstructured failure logs across failure categories; designed annotation schema and multi-stage quality validation pipeline for an ML failure-classification assistant that improved classification accuracy by **15%**, reducing manual triage escalations.
- Developed Python automation scripts for data collection, feature extraction, and preprocessing across the Switch Data Monitoring platform; standardized input formats across heterogeneous device fleets, directly reducing noise in downstream model inputs.

Software Engineer (Graduate Intern)

Jan. 2022 – Jul. 2022

Hewlett Packard Enterprise

Bangalore, India

- Built automated feature validation scripts integrated into CI pipeline and developed data collection scripts for the Switch Data Monitoring platform; eliminated production rejections and reduced pre-release debugging time.
- Contributed bug fixes across the codebase and maintained technical documentation to support team onboarding and platform observability.

PROJECTS

Real-Time Market Anomaly Detection Pipeline | *Kafka, Spark, Delta Lake, GCS, BigQuery, dbt, Docker*

- Built end-to-end real-time pipeline ingesting equity trade events via Aiven Kafka (SSL/TLS, ticker-keyed partitioning) into PySpark Structured Streaming; **1-min** tumbling VWAP windows with **30s** watermarking and fault-tolerant GCS checkpointing — sub-**60s** anomaly detection latency across **5** instruments.
- Implemented a medallion lakehouse on GCS (Bronze raw ticks → Silver aggregations → Gold anomaly events); **3** detection signals — Z-score volume spike, intra-window price deviation, wash trading — each with per-signal confidence scores; Gold loaded to BigQuery; **3** dbt models with **8** passing data quality tests.

Multi-Source Music Data Pipeline | *PySpark, dbt, Airflow, DuckDB, Parquet, SQL*

- Built a batch ELT pipeline processing **46M+** records from 3 sources using medallion architecture; **60%** storage compression via partitioned Parquet; Bronze → Silver → Gold DAG orchestrated in Airflow.
- Implemented star schema with **4** Gold tables; Silver-layer quality checks removed **5,043** duplicates across a **36.7%** cross-source match rate; enabled genre trend analysis and segmentation across **70+** countries.

Credit Risk Strategy Engine: Revenue Recovery | *Scikit-learn, Pandas, Statsmodels, A/B Testing, SHAP*

- Conducted hypothesis-driven EDA on **113K+** loans using t-tests and chi-square tests; Kaplan-Meier survival analysis revealed **70%** of defaults show distress signals within 12 months.
- Trained HistGradientBoosting on **14** bureau features; counterfactual A/B framework recovered **\$7.8M** in loan volume and reduced default rate by **20%**; FCRA-compliant Adverse Action codes via SHAP.